Neural networks LAB1

Neural networks for regression

The goal of this LAB is to understand how neural networks function. We'll use a simple regression task to this end.

Part 1 – Dataset

At its core, a neural network is nothing more than a (very) complicated numerical function whose parameters we need to tune. First, we'll look at the easiest type of numerical function there is : functions defined with 1D quantitative input and 1d quantitative output.

1. Load the two databases for this LAB. How many samples are there in each case ? For both databases, separate the observations and the labels : the first column is the unique feature of this dataset, and the second one is our target.
2. Plot both datasets on two separate graphs (each time, the x-axis must be the feature and the y-axis must be the target).
3. In one of these two cases, the use of a multi-layer perceptron is not recommended. Find which one and why. What better suited model could you use in that case ?

Part 2 – Visualization

4. Make a function that takes a model, a list of 1D-samples (X) and a list of true targets (y_true) as input, and displays a graph where :

- the real values of each sample are shown in green
- the predicted values of each sample are shown in gray

During question 5, focus on the dataset for which neural networks are **not** suited.

5. Train a linear model on the dataset, then show the graph using the function you created in question 4 (you may both train and display using the entire dataset). Conclude.

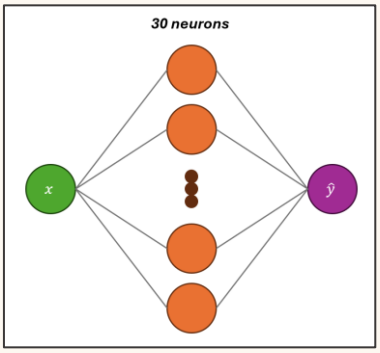
For all following questions in this LAB, focus on the dataset for which neural networks are well-suited.

6. Implement a neural network with the following characteristics :

- one hidden layer with 30 neurons
- a ReLU activation layer
- adaptive learning rate
- a fixed random state for reproducibility
- the maximum amount of iterations at ten thousand (at least !)
- warm start

7. Train the neural network on all the data then display its results using the function you created in question 4. Interpret the results.

8. How many weights and biases does this network have ? Here is a diagram of the neural network to help you :



Part 3 – Training process

In this section, we are going to keep the same neural network, but we will observe the impact of changes in the optimization process, i.e. the way the network learns.
If you don't know about all the optimization methods we are about to cover, make sure to refer to this article and pay attention to its animations :
https://towardsdatascience.com/a-visual-explanation-of-gradient-descent-methods-momentum-adagrad-rmsprop-adam-f898b102325c

9. Get the loss curve of your previous model and show it.

10. Take the same neural network (one hidden layer, 30 neurons). On a single graph, show the loss curves with the following optimization processes :

- one with an SGD solver with constant learning rate and low momentum (0.3)
- one with an SGD solver with constant learning rate and default momentum (0.9)
- one with an SGD solver with constant learning rate and high momentum (0.99)
- one with an SGD solver with constant learning rate and very high momentum (0.999)
- one with an Adam solver

For each model, set the amount of iterations with no change to 1,000, the optimization tolerance to 1e-5 or less, and the maximum amount of iterations to 10,000. This will allow the models to keep training even if the solutions seem to get worse for a time.
Leave all other parameters unchanged.

11. Display the results of the model with low momentum SGD solver using the function you created in question 4. Explain the results.

12. Explain the learning curve of the SGD solver with very high momentum.

13. Which optimization process turned out to be the best one in our case?

Part 4 – Selecting the best architecture

In this section, we will select the best amount of neurons for our network.

14. Split the dataset into a training set and a testing set (the testing set must contain 20% of the overall set).

15. Make ten different neural networks, each with one hidden layer containing 4 neurons and different random states. What does the random state impact? Train all the networks on the training data, and print their testing score. Explain the sentence: "Neural networks are unstable".

16. Make a function that takes an amount of neurons as input, trains 20 identical networks with this amount of neurons (but different random states), and returns the mean score and the standard deviation of the score.

17. Apply this function to all possible amounts of neurons between 2 and 30. Then, display a graph that shows the evolution of the mean score and its standard deviation, relative to the amount of neurons. Use an error bar graph.

18. How many neurons would you select to make the best model for this dataset and why?